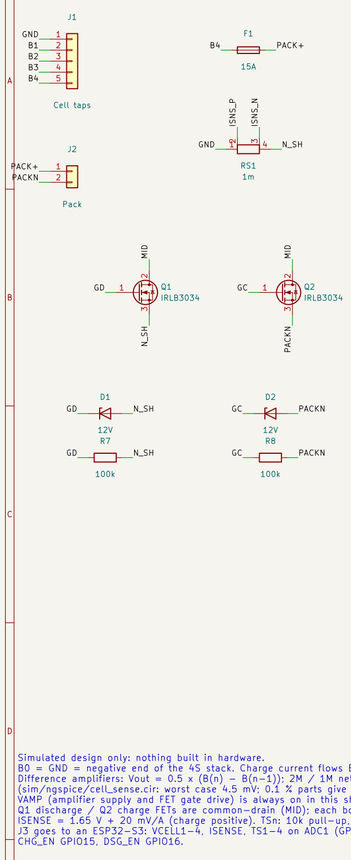
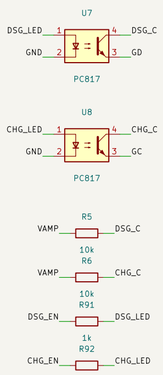
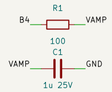
Cell taps, pack terminals, power path



FET drivers (opto, gate clamp)



Amplifier supply



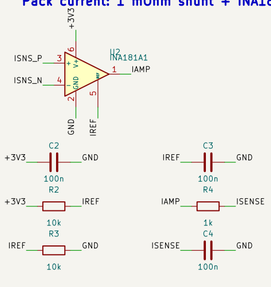
Power flags (ERC)



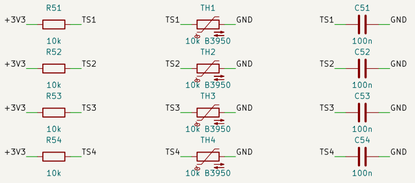
Cell voltage: difference amplifiers, gain 0.5



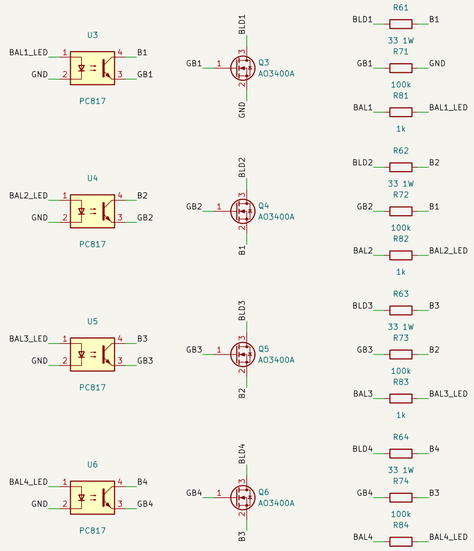
Pack current: 1 mOhm shunt + INA181A1



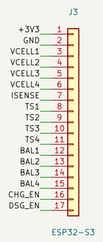
Cell temperature: NTC dividers



Passive balancing: 33 ohm bleed per cell



ESP32-S3 header



Simulated design only: nothing built in hardware.
B0 = GND = negative end of the 4S stack. Charge current flows B0 -> RS1 -> N_SH.
Difference amplifiers: Vout = 0.5 x (B(n) - B(n-1)); 2M / 1M networks matched to 0.01 %
(sim/ngspice/cell_sense.cir: worst case 4.5 mV; 0.1 % parts give 37.7 mV).
VAMP (amplifier supply and FET gate drive) is always on in this sheet: 4 mA from the stack.
Q1 discharge / Q2 charge FETs are common-drain (MID); each body diode passes the other direction.
ISENSE = 1.65 V + 20 mV/A (charge positive). TSn: 10k pull-up, 10k B3950 NTC to GND.
J3 goes to an ESP32-S3: VCELL1-4, ISENSE, TS1-4 on ADC1 (GPIO1-9), BAL1-4 GPIO11-14,
CHG_EN GPIO15, DSG_EN GPIO16.

U1 OPA4197 difference amplifiers, U2 INA181A1, PC817 opto FET and bleed drive
Simulated design only: nothing built in hardware

Shon Zinger

Sheet: /
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Title: 04 — Smart BMS, 4S Li-ion: analog front end and power path (simulat

Size: A2	Date: 2026-11-03	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1